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Context

Design choices, trilateration, and activity recognition

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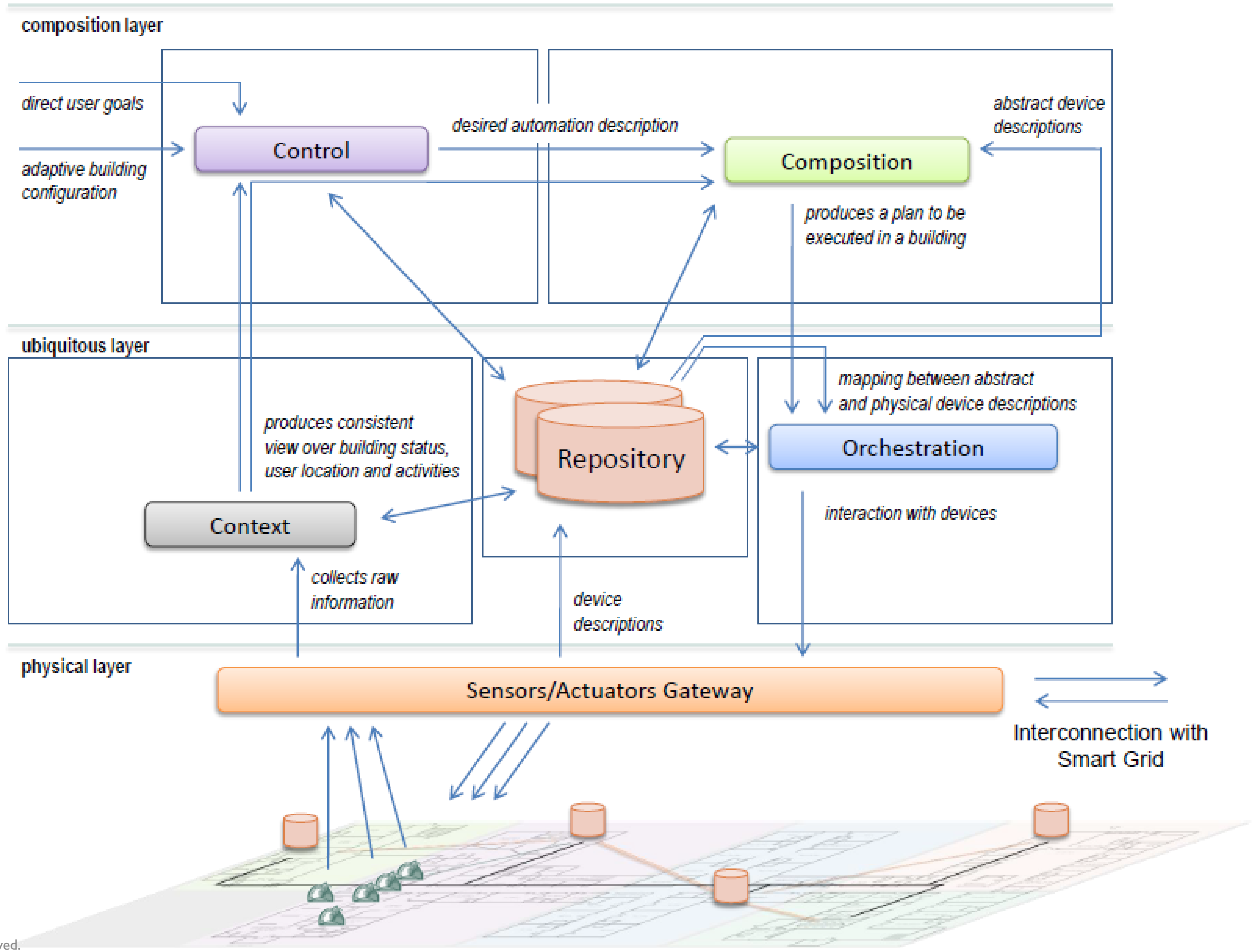
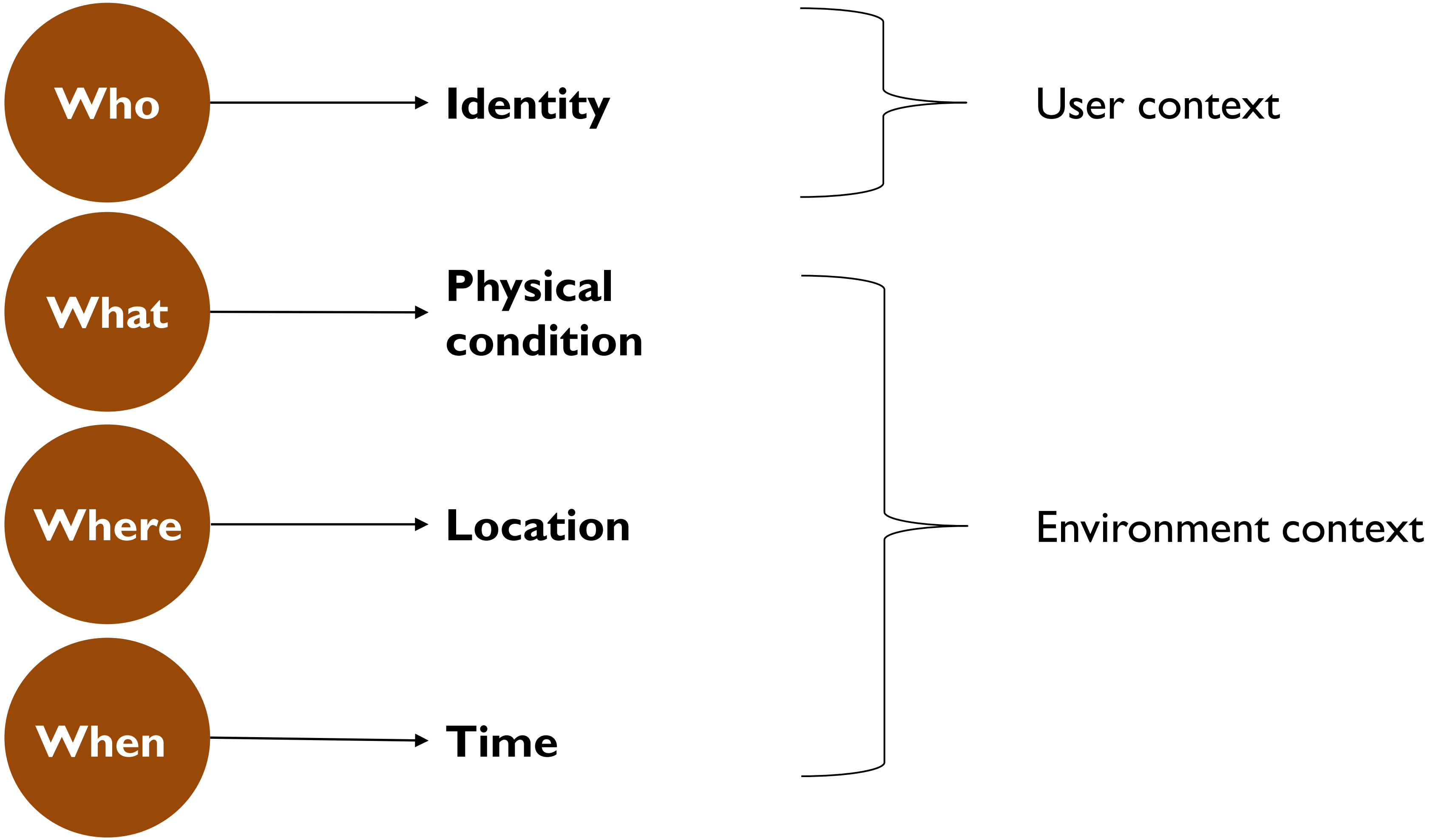

composition layer
direct user goals
adaptive building configuration
Control
desired automation description
Composition
abstract device descriptions
produces a plan to be executed in a building
ubiquitous layer
produces consistent view over building status, user location and activities
Repository
mapping between abstract and physical device descriptions
Orchestration
Context
interaction with devices
collects raw information
device descriptions
physical layer
Sensors/Actuators Gateway
Interconnection with Smart Grid

Context via feature space

- Context related to human factors
  - Information about a person
    - Daily behavioural patterns
    - Emotional state (e.g., angry, calm)
    - Physiological state (e.g., heart rate)
  - Person's social environment
    - Co-location of others
    - Social interaction
    - Group dynamics
  - Person's activities
    - Spontaneous activity
    - Engaged activities
- Context related to physical environment
  - Location
    - Absolute position
    - Relative position
    - Co-location
  - Physical conditions
    - Temperature
    - Air quality

Context via dimensions



Relevance of entities

An entity that is considered *relevant to the interaction* between a user and the system

Example

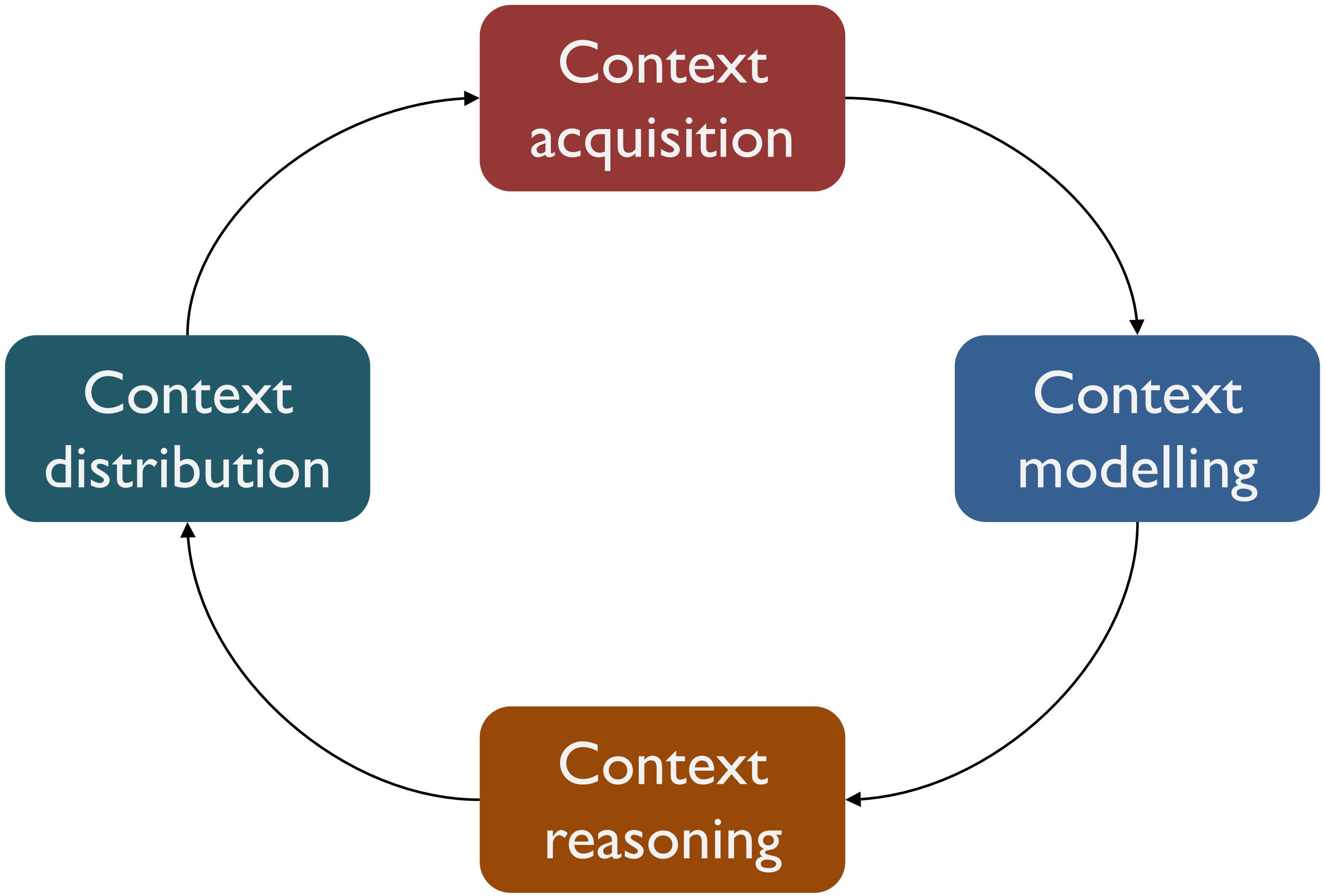
- Consider a smart system to assist museum visitors
- Context could be *a museum visitor is interested in an exhibition item*
- Relevant entities are relative position and native language of museum visitors
- Other entities, such as body temperature, are not relevant

Context awareness

- A system is *context aware* if it uses context to provide relevant information and services to people
- A context-aware system adapts its operation based on context relevant to the environment or people

Where and how to integrate context in a system?

Life cycle of context



Context acquisition

- Environment context
  - Electronic-based IoT
  - Software-based IoT
  - Human-based IoT
  - Virtual IoT
- User context
  - Same types of IoT as for environment context
  - Direct interaction

Examples

Context	Relevant data sources
Temperature	Temperature sensor
Ambient sound	Microphones
Orientation	Compass, magnetic field sensor
Air quality	CO_2 sensor
Energy consumption	Smart meter
Location outdoors	GPS
Location indoors	RFID, Wi-Fi probes, beacons
Identity	Email, RFID
Activity	Accelerometers, PIR motion sensor, pressure sensor, etc.

Design and implementation considerations

- Communication
- Configuration
  - Sensor calibration
  - Frequency
  - …
- Data processing
  - Filtering
  - Aggregation
  - …

Context modelling

- Context information definition and organisation
- Context validation and storage

Modelling techniques

- Key-value pairs
  - Data structure: simple and flat
  - Access: linear search
  - Pros: easy to manage and parse
  - Cons: exact matches, inexpressive structure, weak formalism
- Mark-up schemes
  - Data structure: hierarchical (e.g., XML)
  - Access: mark-up query language
  - Pros: distributed model, W3C standards
  - Cons: weak formalism
- Graphical modelling
  - Data structure: graph (e.g., UML)
  - Access: transformation algorithm
  - Pros: more expressive than key-value and mark-up schemes
  - Cons: lack of formalism for online automated access

Modelling techniques

- Object-oriented modelling
  - Data structure: programming objects
  - Access: accessor methods
  - Pros: distributed OO is mature, reusability
  - Cons: handling incompleteness
- Logic-based modelling
  - Data structure: logical conditions
  - Access: logical reasoning
  - Pros: strong formalism, expressive structuring
  - Cons: handling uncertainty, heterogeneity and temporal aspects; high complexity
- Ontology-based modelling
  - Data structure: concepts and axioms (logic, rules)
  - Access: ontological reasoning
  - Pros: expressive structuring, managing heterogeneity
  - Cons: handling uncertainty, scalability issues

Context reasoning

- Context creation
  - Context generated by processing raw sensor data
    - Scaling
    - Transforming
    - Fusion
    - …
  - Context generated by reasoning over raw sensor data
    - Inference
    - Abstraction
  - Context derived from other contexts
    - Combination or fusion


LIVE
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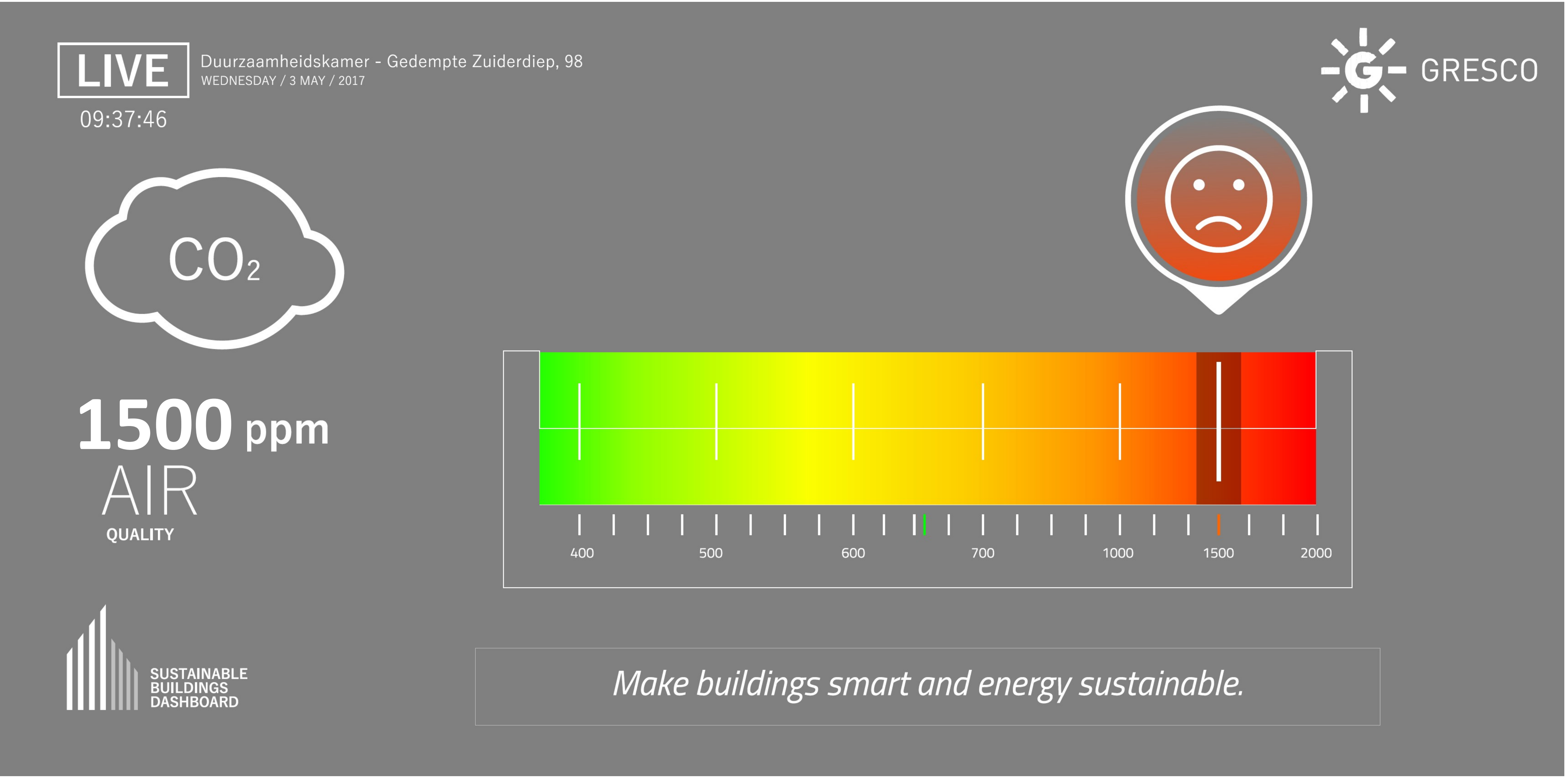
SUSTAINABLE
BUILDINGS
DASHBOARD
Make buildings smart and energy sustainable.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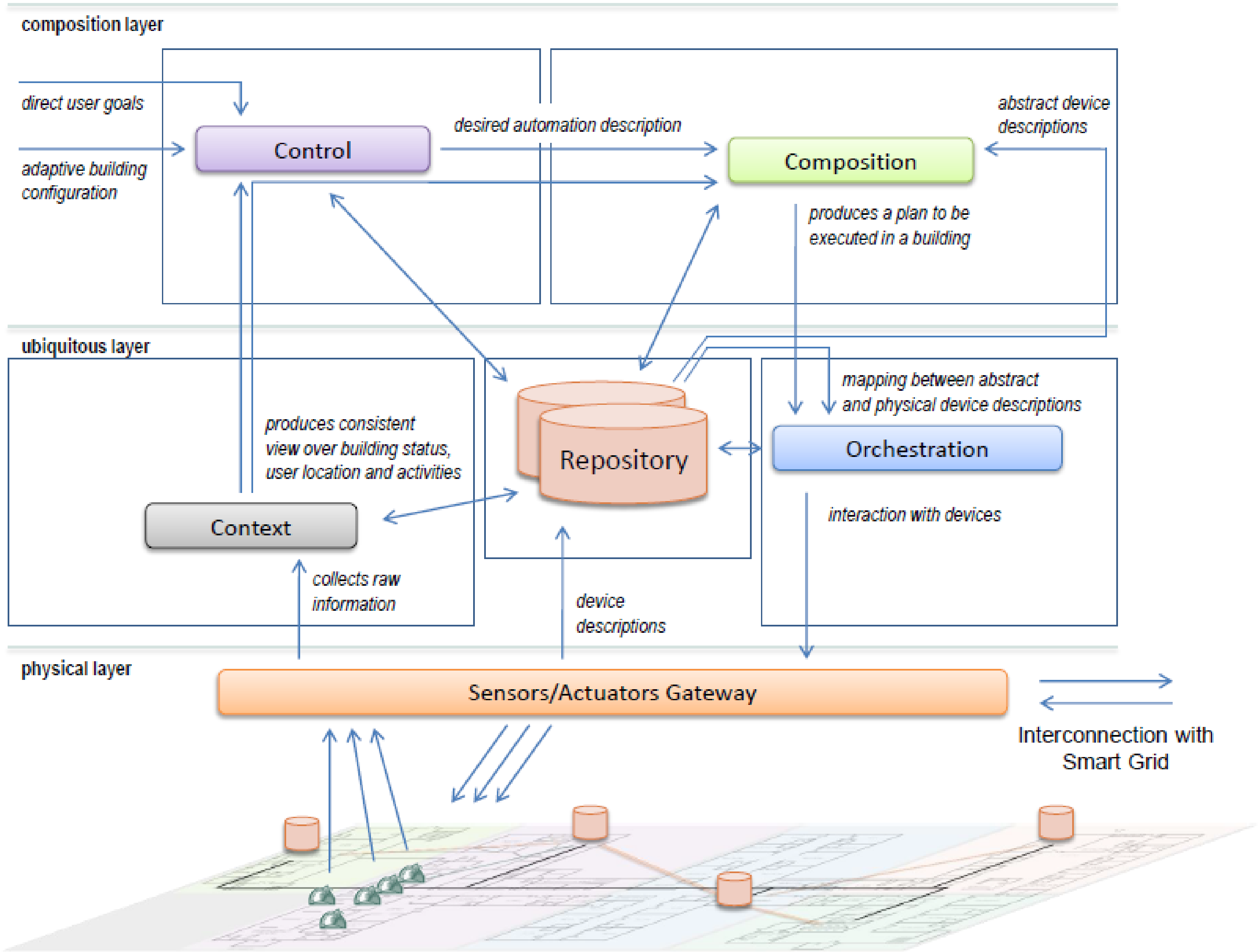
Reasoning techniques

- Supervised learning
- Unsupervised learning
- Rule-based engines
- Fuzzy logic
- Ontological reasoning
- Probabilistic reasoning

Context distribution

- Request-reply communication
- Indirect communication

Adapt to context



Exercises

- Describe why context is relevant for smart systems, and detail two scenarios where smart systems can be used within an elderly care environment.
- Consider the need for a smart system to adapt to context automatically. Give an example of an application in which the high speed at which such a system can adapt is good and another example in which high speed is not good.
- What aspects are important to consider when designing and implementing a context component for smart systems?
- Give three examples of context-aware behaviour of a smart system for controlling the heating system of a hospital.

TRILATERATION

Problem

Given a representation of physical space, a set of sensor readings, and an object of interest, assign a location to the object in the representation of physical space which best represents its actual physical presence.

Localisation techniques

- Range-based localisation
  - Triangulation
  - Trilateration
- Ranging techniques for trilateration
  - Received Signal Strength Indicator (RSSI)
  - Time of Arrival (ToA)

Trilateration (1)

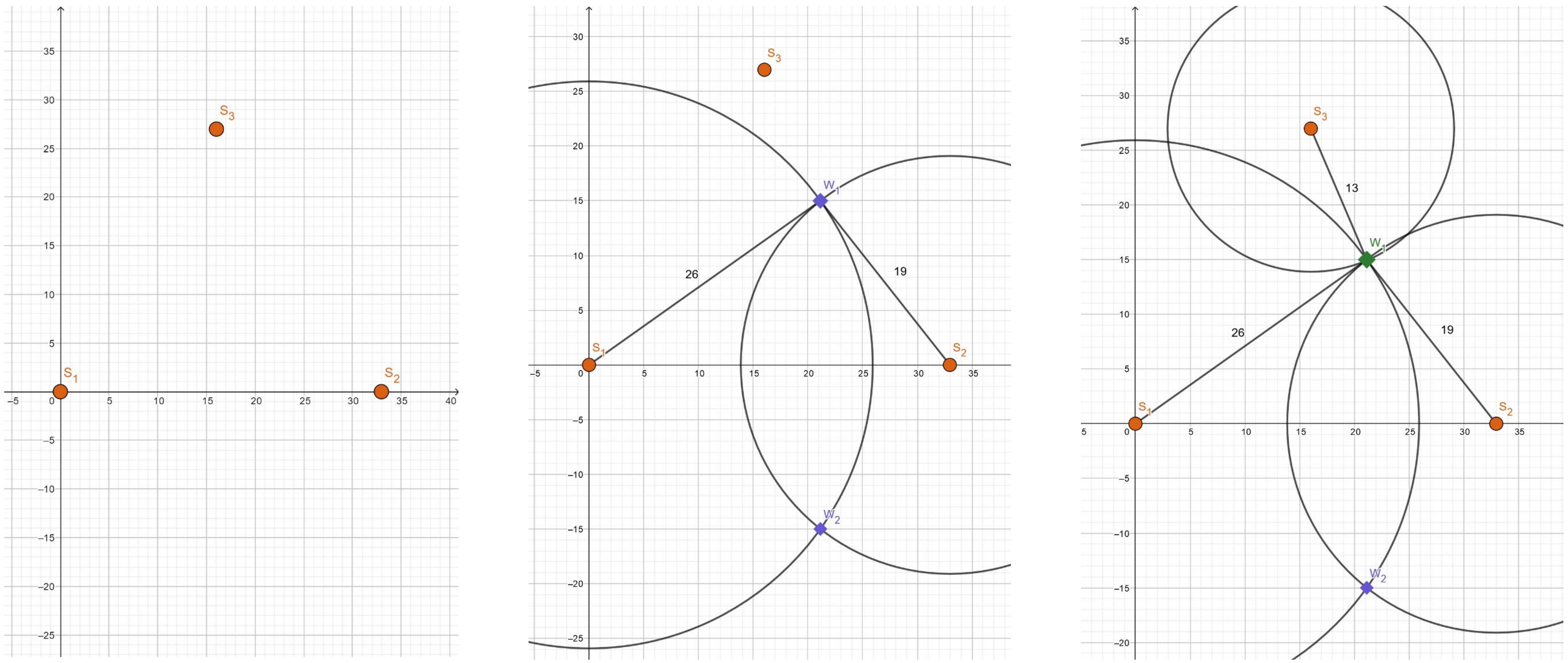
- Localisation based on measured distances between a node and several anchor points with known locations
- Basic concept
  - Given the distance of an anchor, it is known that the node must be along the circumference of a circle centered at anchor and a radius equal to the node-anchor distance
- Needed anchors
  - Two-dimensional space: at least three non-colinear anchors
  - Three-dimensional space: at least four non-coplanar anchors

Trilateration (2)

- Given
  - (x_i, y_i), which are coordinates of an anchor point i
  - r_i, which is a distance between anchor point i and a node u
- Needed
  - (x_u, y_u), which are coordinates of a node u
- Solution
  - Graphical approach
  - Matrix equation
  - System of equations

Exercise

Consider three GPS satellites, S_1, S_2, and S_3, and one smartwatch W. Suppose that, in a rectangular coordinate system, the locations of the satellites S_1, S_2, and S_3 are (0,0), (33,0), and (16,27), respectively. Note that 1 unit represents 1 km. The distances between W and S_1, S_2, and S_3 are 26 km, 19 km, and 13 km, respectively. Assume that S_1, S_2, S_3 and W lie on the same horizontal plane (two-dimensional space). Find the location of the smartwatch W using trilateration.


S1
S2
S3
W1
W2
26
19
13

Matrix equation

$$2\begin{bmatrix} x_3 - x_1 & y_3 - y_1 \\ x_3 - x_2 & y_3 - y_2 \end{bmatrix}\begin{bmatrix} x_u \\ y_u \end{bmatrix} = \begin{bmatrix} (r_1^2 - r_3^2) - (x_1^2 - x_3^2) - (y_1^2 - y_3^2) \\ (r_2^2 - r_3^2) - (x_2^2 - x_3^2) - (y_2^2 - y_3^2) \end{bmatrix}$$

$$(x_1, y_1) = (0,0)$$
$$(x_2, y_2) = (33,0)$$
$$(x_3, y_3) = (16,27)$$

$$r_1 = 26$$
$$r_2 = 19$$
$$r_3 = 13$$

$$2\begin{bmatrix} 16 & 27 \\ -17 & 27 \end{bmatrix}\begin{bmatrix} x_u \\ y_u \end{bmatrix} = \begin{bmatrix} 1492 \\ 88 \end{bmatrix}$$

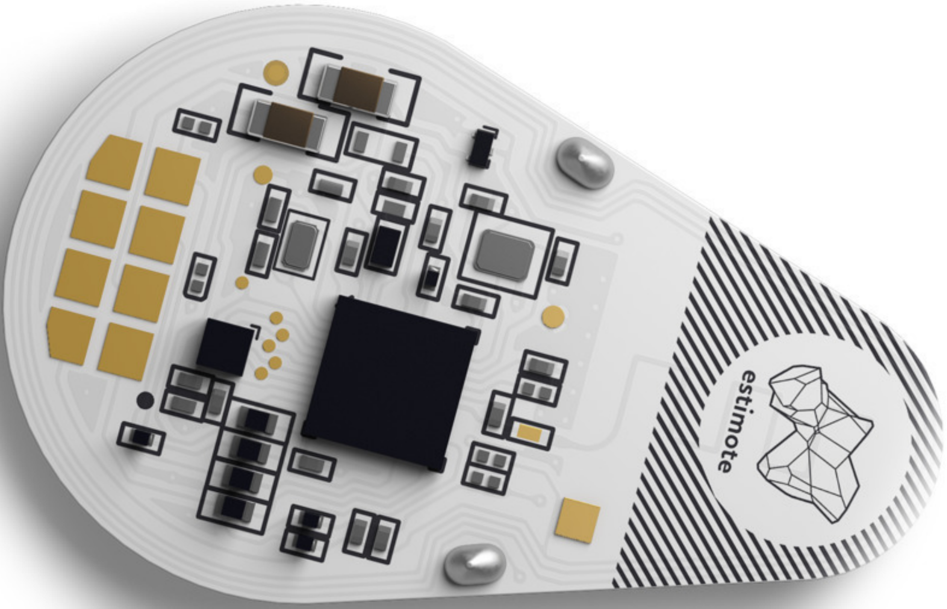
$$\begin{cases} 32x_u + 54y_u = 1492 \\ -34x_u + 54y_u = 88 \end{cases}$$

$$\begin{cases} x_u \cong 21 \\ y_u \cong 15 \end{cases}$$

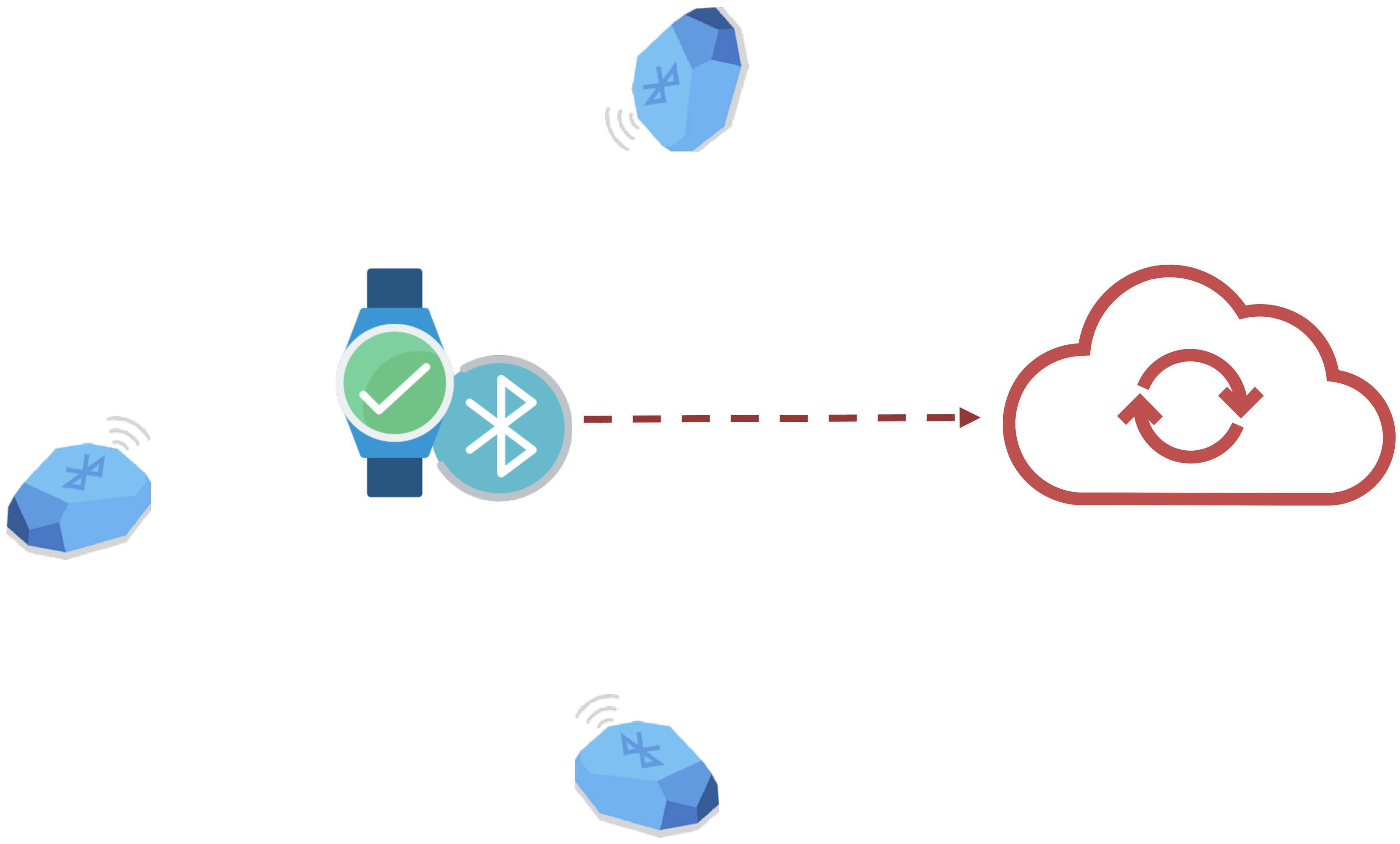
Exercise

Consider three GPS satellites, S_1, S_2, and S_3, and one smartwatch W. Suppose that, in a rectangular coordinate system, the locations of the satellites S_1, S_2, and S_3 are (0,0), (21,0), and (9,15), respectively. Note that 1 unit represents 1 km. The distances between W and S_1, S_2, and S_3 are 13 km, 20 km, and 5 km, respectively. Assume that S_1, S_2, S_3 and W lie on the same horizontal plane (two-dimensional space). Find the location of the smartwatch W using trilateration. Does W lie within or outside $\Delta S_1 S_2 S_3$?

Beacons



Indoor location using beacons



Estimote Proximity Beacons

- Default settings
  - Broadcasting Power: -4 dBm
  - RSSI: [-26,-100] at maximum broadcasting power
  - Measured Power: -66
- Advertising packet types
  - Estimote Location
  - Estimote Telemetry

Algorithm

- Scan for Estimote Proximity beacons
- Upon a beacon's detection, calculate a distance estimation
- When three beacons have been detected and their distances are known, calculate approximate position using trilateration

Python library

- `beacontools`
  - Parsing and scanning library
  - Supports Eddystone, iBeacon and Estimote beacons
  - For Estimote beacons, it supports only advertising packets of type Estimote Telemetry

Beacons storage

```
beacons = {
  "beacon_1": {
    "x": 0,
    "y": 0,
    "r": 0
  },
  "beacon_2": {
    "x": 7,
    "y": 3,
    "r": 0
  },
  "beacon_3": {
    "x": 4,
    "y": 8,
    "r": 0
  }
}
```

Scanning

```python
import time
from beacontools import BeaconScanner


def callback(bt_addr, rssi, packet, additional_info):
    print("<%s, %d> %s %s" % (bt_addr, rssi, packet, additional_info))
    # needs to be revised to account for distance estimation of all three beacons and
    # for position estimation once distances of all three beacons are known


# scan for any packet
scanner = BeaconScanner(callback)
scanner.start()

time.sleep(30)
scanner.stop()
```

Distance estimation

$$d = \sqrt[\alpha]{\frac{cP_{tx}}{Precv}}$$

- Assume
  - $P_{tx} = -66$
  - $c = 2$
  - $\alpha = 2$

$$d = \sqrt{\frac{2P_{tx}}{Precv}}$$

Table 3.6 Path loss exponent and standard deviation measured in different buildings [And94]

Building	Frequency (MHz)	n	σ (dB)
Retail Stores	914	2.2	8.7
Grocery Store	914	1.8	5.2
Office, hard partition	1500	3.0	7.0
Office, soft partition	900	2.4	9.6
Office, soft partition	1900	2.6	14.1
Factory LOS			
Textile/Chemical	1300	2.0	3.0
Textile/Chemical	4000	2.1	7.0
Paper/Cereals	1300	1.8	6.0
Metalworking	1300	1.6	5.8
Suburban Home			
Indoor Street	900	3.0	7.0
Factory OBS			
Textile/Chemical	4000	2.1	9.7
Metalworking	1300	3.3	6.8

Rappaport, T. Wireless Communications: Principles & Practice, Prentice Hall, Inc., 2002.

Distance estimation

```
def estimate_distance(rssi):
    p_tx = -66
    c = 2
    return math.sqrt(c*p_tx/rssi)
```

Position estimation

$$\underbrace{2(x_3 - x_1)}_{a} x_u + \underbrace{2(y_3 - y_1)}_{b} y_u = \underbrace{(r_1^2 - r_3^2) - (x_1^2 - x_3^2) - (y_1^2 - y_3^2)}_{c}$$

$$\underbrace{2(x_3 - x_2)}_{d} x_u + \underbrace{2(y_3 - y_2)}_{e} y_u = \underbrace{(r_2^2 - r_3^2) - (x_2^2 - x_3^2) - (y_2^2 - y_3^2)}_{f}$$

$$ax_u + by_u = c$$

$$dx_u + ey_u = f$$

$$x_u = \frac{ce - bf}{ae - bd} \qquad y_u = \frac{af - cd}{ae - bd}$$

Position estimation

```
def estimate_position():
    a = 2 * (beacons["beacon_3"]["x"] - beacons["beacon_1"]["x"])
    b = 2 * (beacons["beacon_3"]["y"] - beacons["beacon_1"]["y"])
    c = beacons["beacon_1"]["r"] ** 2 - beacons["beacon_3"]["r"] ** 2 - beacons["beacon_1"]["x"] ** 2 + \
        beacons["beacon_3"]["x"] ** 2 - beacons["beacon_1"]["y"] ** 2 + beacons["beacon_3"]["y"] ** 2
    d = 2 * (beacons["beacon_3"]["x"] - beacons["beacon_2"]["x"])
    e = 2 * (beacons["beacon_3"]["y"] - beacons["beacon_2"]["y"])
    f = beacons["beacon_2"]["r"] ** 2 - beacons["beacon_3"]["r"] ** 2 - beacons["beacon_2"]["x"] ** 2 + \
        beacons["beacon_3"]["x"] ** 2 - beacons["beacon_2"]["y"] ** 2 + beacons["beacon_3"]["y"] ** 2
    x = (c * e - b * f) / (a * e - b * d)
    y = (a * f - c * d) / (a * e - b * d)
    return x, y
```

ACTIVITY RECOGNITION

Problem

Given a location, a set of sensor readings, and a person, recognise the physical activity the person is performing in the location

Activity recognition

- Types of sensors
  - Vision sensors
  - Wearable sensors
  - Simple sensors
- Types of techniques
  - Based on machine learning
  - Based on logical reasoning
- Types of environments
  - Elderly care homes
  - Households
  - Offices
  - …

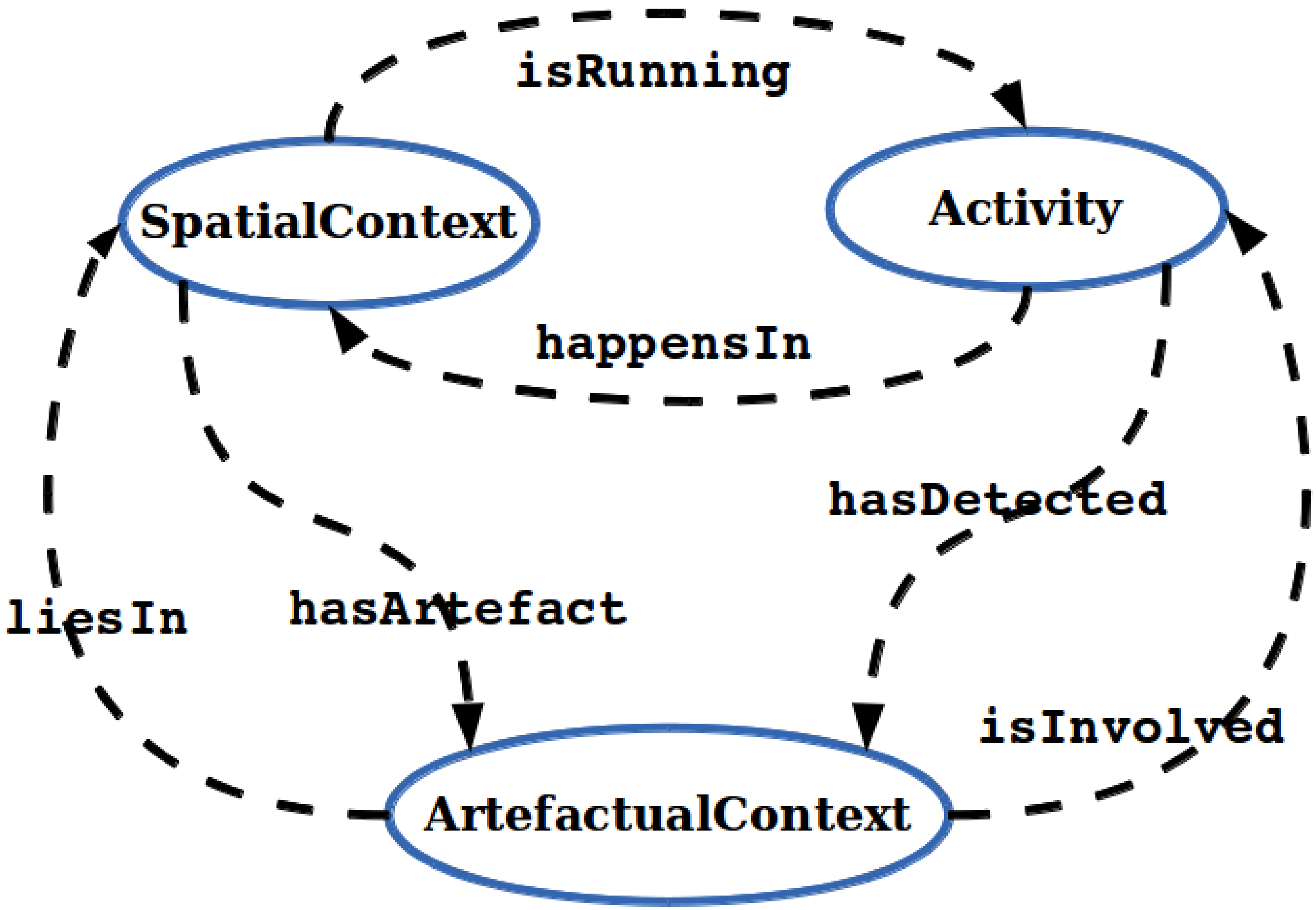
Closer look

- Simple sensors
- Ontological reasoning
- Office building

Office activities

Office AA	Performed activities	Definition	Involved artefacts
WorkingRoom	Working with PC	User is using PC, PC and monitor are turn on	Chair, PC, and monitor
	Working without PC	User is sitting at the desk but PC and monitor are turned off	Chair
	Presence	User is active in the room but no further specific activity is recognized	Movement detector
	Absence	User is absent from the room	No artefacts involved
MeetingRoom	Giving a presentation	User is giving a presentation with the projector	Chairs, Projector, human voice detector
	Having a meeting	Users are discussing at the meeting table	Chairs, human voice detector
	Presence	User is active in the room but the specific activity is not recognized	Movement detector
	Absence	User is absent from the room	No artefacts involved
CoffeeCorner	Having a coffee break	Users are having lunch/coffee or just a break	Coffee machine, microwave, movement detector
	Absence	There is no one at the CoffeeCorner	No artefacts involved

Overview of ontologies



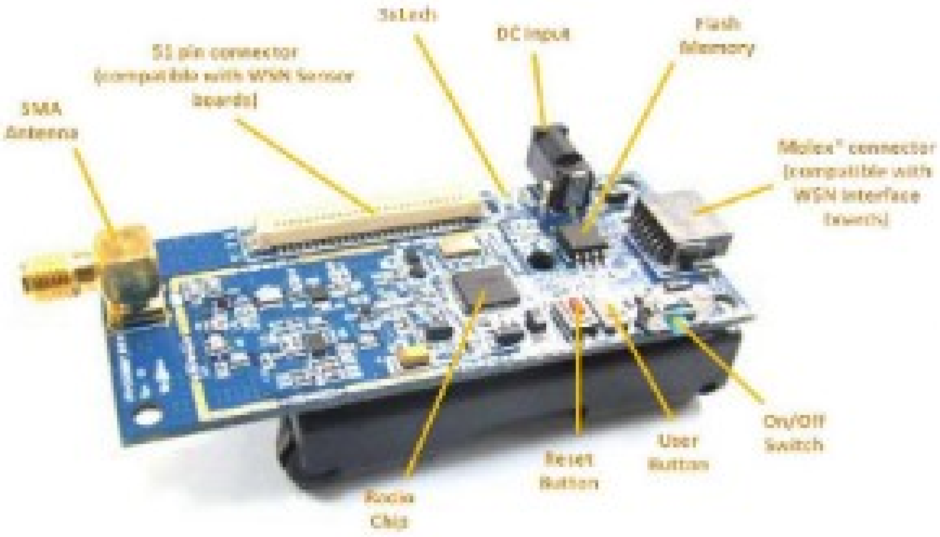
Algorithm

- Get the current readings of all relevant sensors
- Convert sensor readings to corresponding properties in the ontologies
- Aggregate and fuse sensor readings using ontologies
- Reason over ontologies to recognise activities performed in all activity areas at the current time
- Return the result of reasoning as a list of activities with corresponding areas

Implementation

- Sensors
  - Passive infrared sensor
  - Pressure sensor
  - Microphone
  - Electricity measuring plug
- Ontologies
  - OWL2 Web Ontology Language
  - Protégé
- Ontological reasoner
  - HermiT
- Algorithm

Sensors

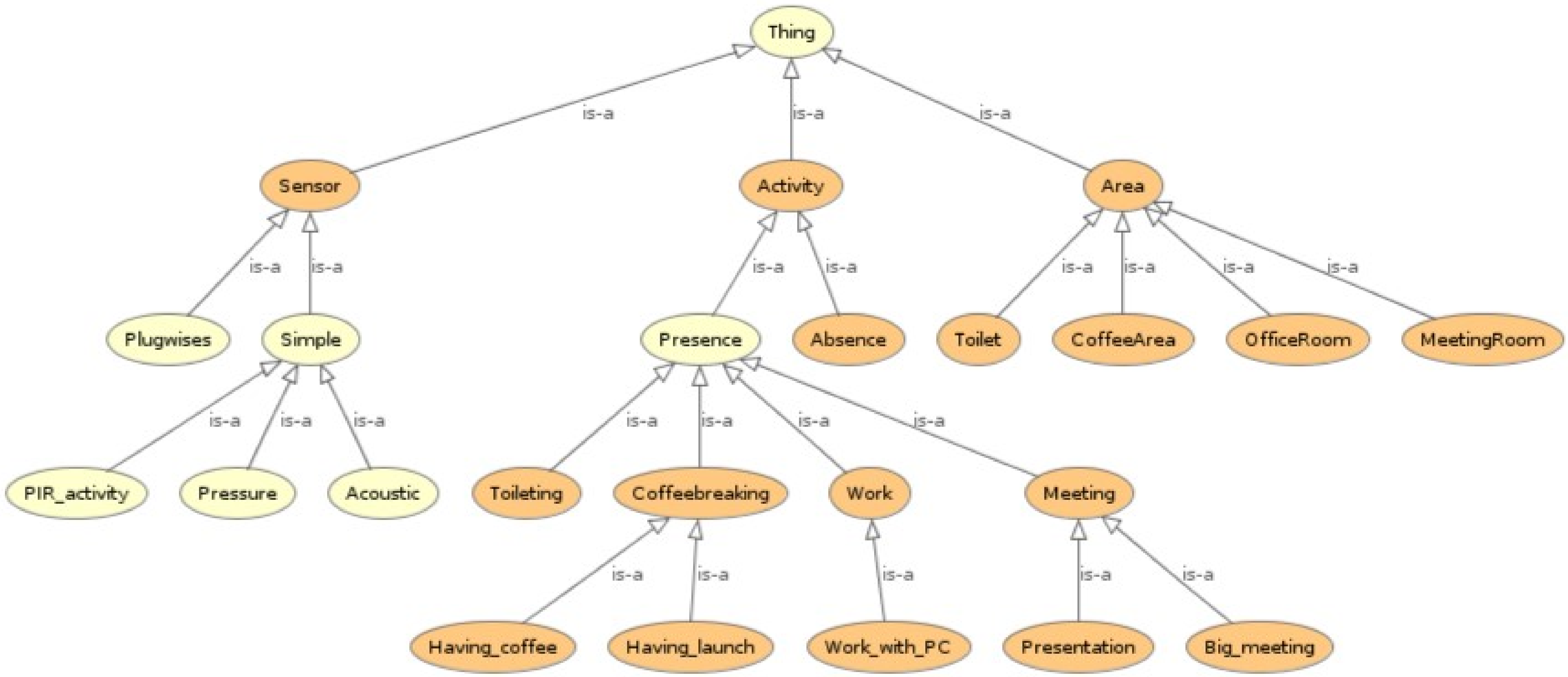


(a) Advantic TelosB mote

(b) Tekscan® A201-100 pressure sensor

(c) PIR and Acoustic sensors

Fig. 8: Plugwise Cirle, Circle+, and Stick


Thing
is-a
is-a
is-a
Sensor
Activity
Area
is-a
is-a
is-a
is-a
is-a
is-a
is-a
is-a
Plugwises
Simple
Presence
Absence
Toilet
CoffeeArea
OfficeRoom
MeetingRoom
is-a
is-a
is-a
is-a
is-a
is-a
is-a
PIR_activity
Pressure
Acoustic
Toileting
Coffeebreaking
Work
Meeting
is-a
is-a
is-a
is-a
is-a
Having_coffee
Having_launch
Work_with_PC
Presentation
Big_meeting

Ontologies in OWL2

- $\#Area$
  - $\#Area \equiv \exists(\#isRunning.\#Activity)$
  - $\#Area \sqsubseteq (\#MeetingRoom \sqcup \#OfficeRoom \sqcup \#CoffeeCorner)$
  - Subclasses
    - $\#CoffeeCorner \sqsubseteq \#Area$
    - $\#CoffeeCorner \equiv \#isRunning.(\#Absence \sqcup \#CoffeeBreak)$

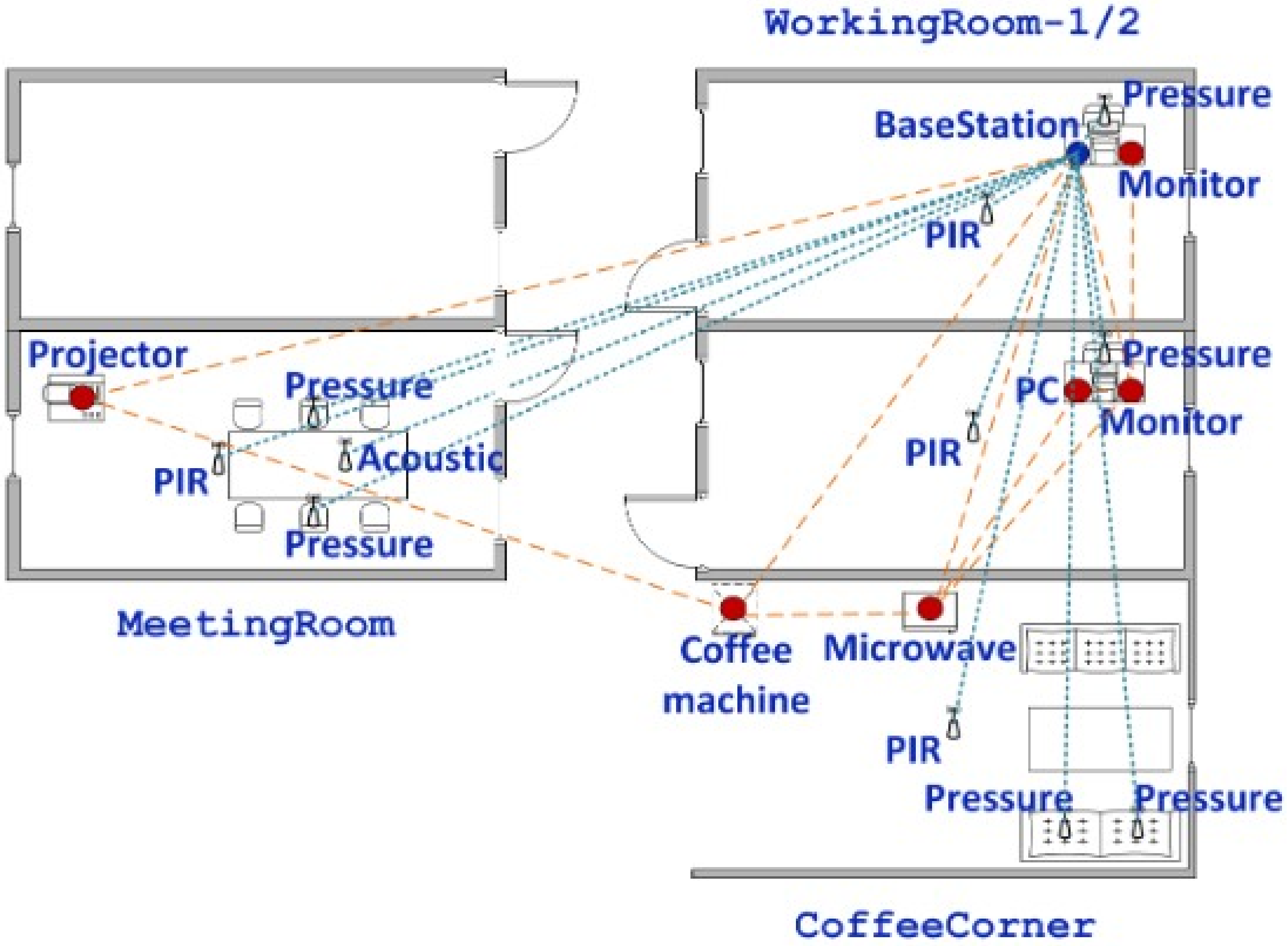
Ontologies in OWL2

- $\#Sensor$
  - $\#Sensor \sqsubseteq (\#Plugwises \sqcup \#Simple)$
  - $\#Sensor \equiv (\exists \#liesIn.\#Area) \sqcap (\#hasID: nonNegativeInteger)$
  - Subclasses (see reference [4])
  - Sensor instances as individuals
    - $\#CoffeeConer_{Acoustic} \in \#Acoustic \sqcap (\#liesIn.\#CoffeeCorner)$
    - $\#CoffeeCorner_{PIR} \in \#PIR \sqcap (\#liesIn.\#CoffeeCorner)$
    - $\#CoffeeCorner_{Microwave} \in \#Plugwise \sqcap (\#liesIn.\#CoffeeCorner)$

Ontologies in OWL2

- $\#Activity$
  - $\#Activity \sqsubseteq (\#Absence \sqcup \#Presence)$
  - $\#Activity \equiv \exists(\#hasDetected.\#Sensor)$
  - Subclasses
    - $\#CoffeeBreak \sqsubseteq \#Activity$
    - $\#CoffeeBreak \equiv \exists\#hasDetected.(\#liesIn{:}\#CoffeeCorner)$

Experimental setup



Ground truth

Table 7.8: Ground truth of the occurrences of activity instance at each activity area

Area	Activity						
	Wpc	W	P	M	C	B	A
WorkingRoom-1	3600	2700	0	0	0	1800	9900
WorkingRoom-2	4500	2700	0	0	0	900	9900
MeetingRoom	0	0	900	3600	0	0	13500
CoffeeCorner	0	0	0	0	2700	0	15300

A = Absence; B = Being present; W = Working without PC; Wpc = Working with PC;M = Having a meeting; P = Giving a presentation; C = Coffee break

Results

Table 7.9: Success rates of the system at each area

Area	$\sum_{\forall activities} FN$	Success rate
`WorkingRoom-1`	1429	92.06%
`WorkingRoom-2`	1403	92.21%
`MeetingRoom`	939	94.78%
`CoffeeCorner`	463	97.43%
$T_{all} = 18000\ seconds$		

Table 7.10: Results for working room 1

(a) Confusion matrix

Recognised as →	Wpc	W	B	A
Working with PC	**3349**	251	0	0
Working without PC	163	**2474**	33	30
Being present	60	187	**1511**	42
Absence	149	30	484	**9237**

(b) Precision/Recall

Activity	Precision (%)	Recall(%)
Working with PC	90.00	93.03
Working without PC	84.09	91.63
Being present	74.51	83.94
Absence	99.23	93.30

A = Absence; B = Being present; W = Working without PC; Wpc = Working with PC

Final remarks

- At the University of Stuttgart, we have experts in AI planning, automated service composition, and smart energy systems, should you be interested in doing a research project, thesis, or PhD.
- We offer both theoretical and application-oriented research projects, although availability may vary based on current capacity.
- I hope you enjoyed the course. Good luck with the project demo and exam.

Thank you for taking the course

References

1. Poslad, S., Ubiquitous Computing: Smart Devices, Environments and Interactions, John Wiley & Sons Ltd., 2009.
2. Perera, C., Zaslavsky, A., Christen, P., Georgakopoulos, D., Context Aware Computing for The Internet of Things: A Survey, IEEE Communications Surveys & Tutorials, 16(1): 414-454, IEEE, 2014.
3. Schmidt, A. Ubiquitous Computing – Computing in Context, PhD thesis, Lancaster University, 2002.
4. Nguyen, T. A., Energy Adaptive Buildings: From sensor data to being aware of users, PhD thesis, University of Groningen, 2015.
5. Aggarwal, J. K. and Ryoo, M. S. 2011. Human activity analysis: A review. ACM Comput. Surv. 43, 3, Article 16.

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